



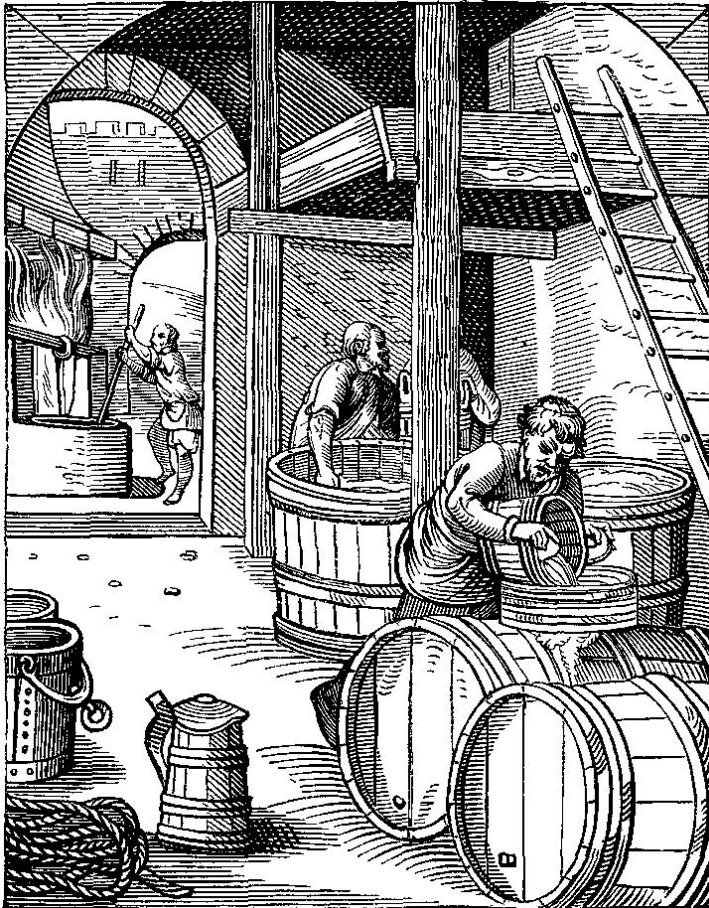
Introduction to Biology BIO 101

Biotechnology & Bioethics Lecture: 11

Biotechnology

- Combination of biology and technology.
- Use of technology to modify or upgrade part or whole of a biological organism or the system for industrial and human welfare. Mostly microorganisms, tissue cells are used.
- Application ranges from **agriculture** to **industry** to **medicine, nutrition, environmental conservation etc.** making it one of the fastest growing fields.

History of Biotechnology



7000 B. C.

Pre-historic biotechnology

- Consumption of wild plants
- Yeast used in making bread, vinegar and beer/wine (6000 B. C.)
- Organized agriculture; crop rotation (3000 B. C.)
- Application of curd molds on wounds (500 B. C.)
- Transfer of features from parents to offspring (Socrates, 420 B. C.)
- Anton Van Leeuwenhoek observes live cells (1673)

1700 A. D.

Historical biotechnology

- Edward Jenner's smallpox vaccine (1796)
- Microbial fermentation by Louis Pasteur (1863)
- Mendel's laws of inheritance (1865)
- Pasteur and Roux develop rabies vaccine (1885)

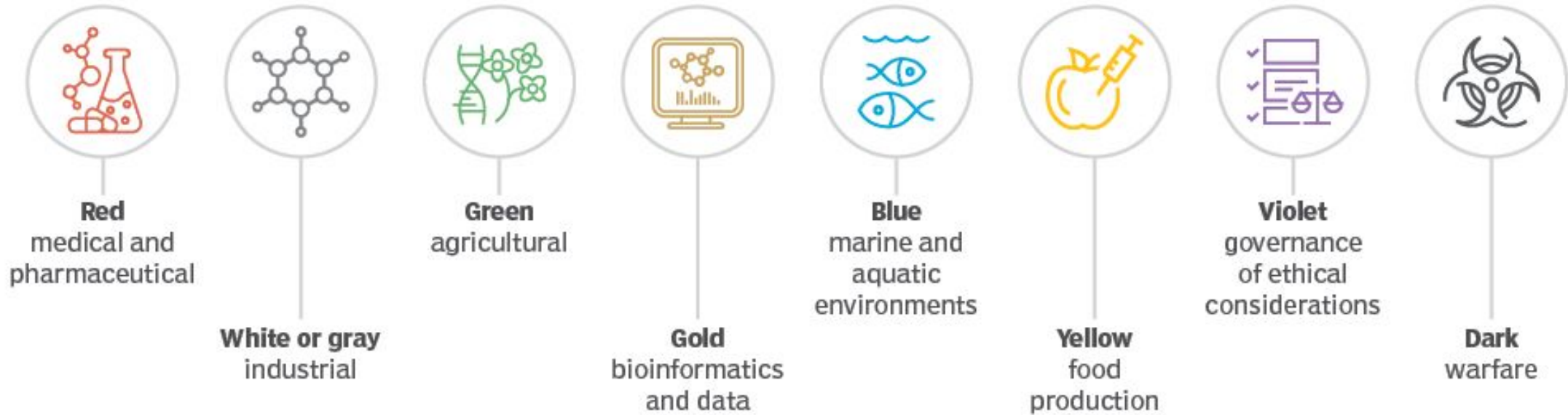
1900 A. D.

Modern biotechnology

- Robert Koch's postulates for pathogenicity (1890)
- Mutation theory by Hugo de Vries (1900)
- Sutton discovers Chromosomes (1902)
- Morgan establishes role of chromosomes in heredity (1907)
- Johannsen coins the word "gene" (1909)
- Griffith reports "transforming principle" as genetic material (1928)
- Flemming finds out Penicillin as an antibiotic (1928)
- Watson and Crick reports DNA double helix (1953)
- Beginning of recombinant DNA technology and genetic engineering (1970s)

Branches of Biotechnology

Color codes are used for the classification of the branches of biotechnology.



Red or Medical and Pharmaceutical Biotechnology

Biotechnology in the medical, pharmaceutical industries and health preservation.

- Production of medicines and pharmaceutical products for treating or diagnosing disorders
- Gene therapy for prevention and control of diseases like hemophilia cystic fibrosis
- Development of vaccines and antibodies for rabies, covid 19, etc.

White or Industrial Biotechnology

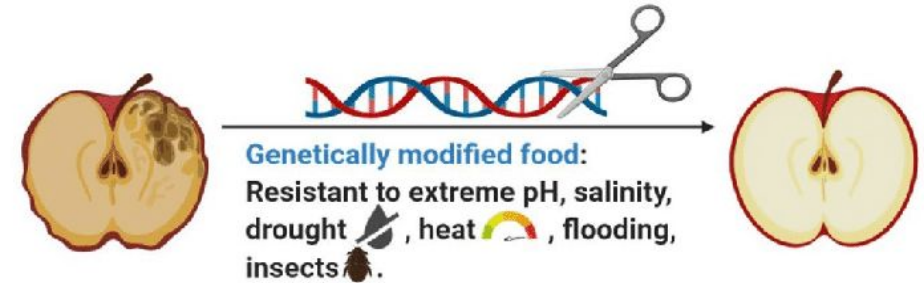
Biotechnology involved in application of biocatalysis in industries.

- Modifying enzymes of microbes that work under extreme conditions
- Biofertilizers and Biopesticides

Green or Agricultural Biotechnology

- Biotechnology involved in improving plant nutritional quality, quantity, and production.

- Crops genetically modified (GM) to create variants that are more resistant to drought, insects, can generate higher yield, emit low CO_2
- Can carry essential components. For e.g. golden rice was genetically modified to produce beta-carotene (a precursor of vitamin A). A diet including golden rice can thus help to raise vitamin A level.



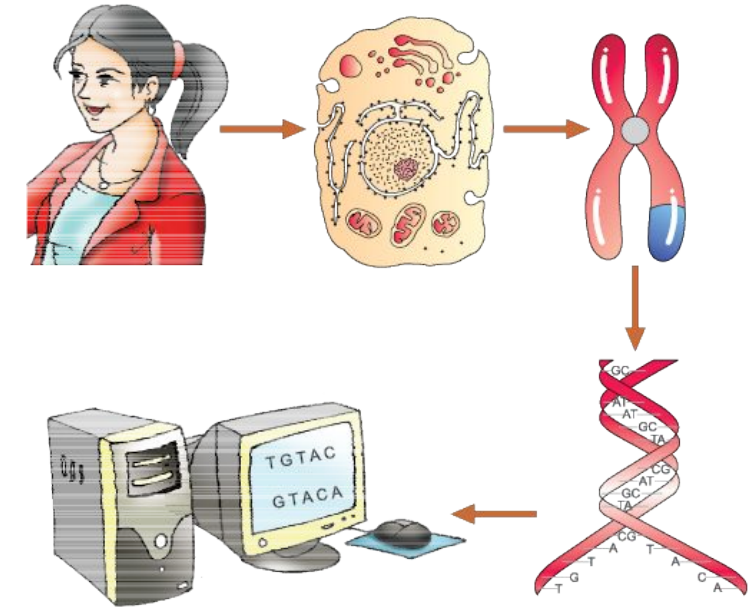
Negative impacts:

- On human health: introducing new allergens and toxins
- On ecosystem: loss of biodiversity, insecticide resistance etc.

Gold or Bioinformatics & Data Biotechnology

Biotechnology that addresses and analyze biological problems using computational techniques. Main areas include:

- **Bioinformatics:** Field that focuses on analyzing large sets of biological data.
- **Nanotechnology:** Field that uses technology at a nanoscale (atomic, molecular etc.)
- **Computational Biology:** Use computational methods to develop models for the study of biological systems.



- The **Human Genome Project (HGP)** was an international scientific research project with the goal of mapping the complete set of nucleotides in human genome (> 3 billion). This was a 13-year project (1990-2003) coordinated by the U.S. Department of Energy and the National Institute of Health. The sequence of the human genome generated by this project was not from a single person. Rather, it reflects a patchwork from multiple people.
- Helps understand & identify ~25,000 human genes which can help develop new ways to diagnose, treat, prevent disorders. Many non-human organisms have also been sequenced (yeast, fruit fly, rice).

Blue or Marine Biotechnology

Biotechnology involved in using science and technology on aquatic and marine organisms to gain knowledge and benefits.

Applications:

- ❑ Preserve aquatic diversity, important for natural stability.
- ❑ GM aquatic organisms can be sources of nutritious food: fish oil (components that can reduce inflammation)
- ❑ Sea weeds (marine algae) can be turned into effective biofuels (emit low level of CO₂).

Problems:

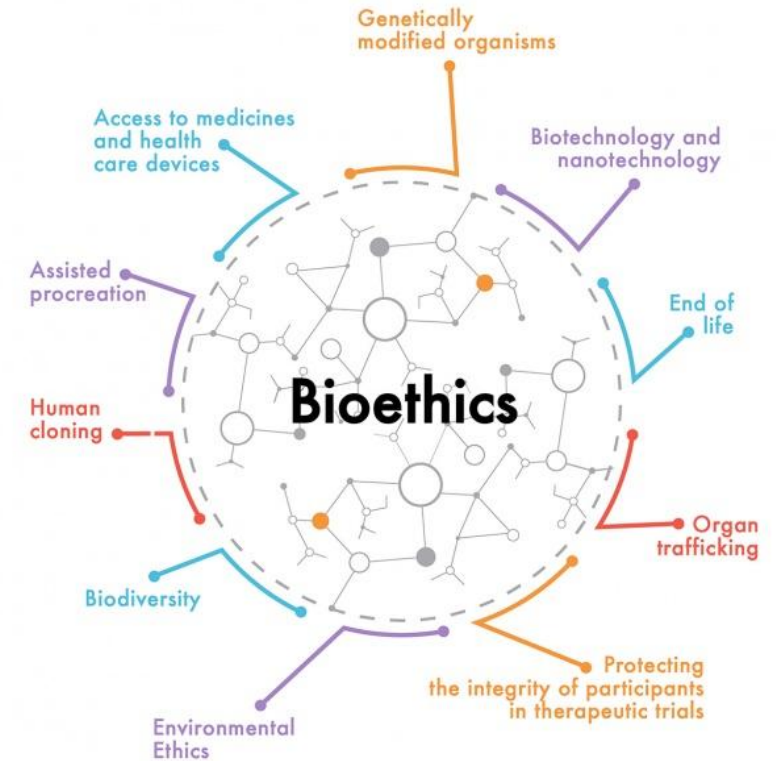
- ❑ Safety concerns about GM food
- ❑ Fishes bred have high level of growth hormone
- ❑ Threat for natural aquatic organisms

Other Biotechnology Fields

- **Yellow or nutritional biotechnology** is dedicated to the food industry to obtain healthier, nutritious, tasty foods by incorporating elements of biological origin: microbes, insects.
- **Grey Biotechnology** emphasizes on the maintenance of biodiversity and removal of pollutants.
- **Purple or legal biotechnology** ensures biotech processes meet ethical and legal obligations.
- **Brown or desert biotechnology** involves the management of soil in arid lands.
- **Black or War Biotechnology** deals with the development of biological weapons for bioterrorism and biowarfare. The most dangerous and unethical as consequences can be catastrophic.

Bioethics

- Bioethics is the study of ethical, social, and legal issues that arise in biomedicine and biomedical research.
- Bioethics includes:
 - Medical Ethics** (issues in health care), **Research Ethics** (issues in the conduct of research), **Environmental Ethics** (issues pertaining to the relationship between human activities and the environment)
- Purpose:
 - Increasing public awareness about bioethical issues, educating them about the principles, informed decision making.
 - Promoting transparency, encourage holding discussions
 - Encouraging responsible reporting and discussions in the media



Case Studies in Biotech Ethics: HeLa Cells and Informed Consent

- Henrietta Lack's cells (HeLa cells) were taken without her consent in 1951 during a medical procedure. The cells became crucial in scientific research due to their unique ability to replicate indefinitely.
- Bioethical issues raised:
 - ❑ **Lack of informed consent:** Henrietta Lacks' cells were obtained without her knowledge or consent, raising ethical concerns about patient autonomy and rights over their biological samples.
 - ❑ **Profit:** HeLa cells' commercialization and widespread distribution in research raised questions about profit derived from biological materials taken without consent.

Henrietta Lacks' story brought attention to historical injustices in medical research and the need for ethical guidelines to protect patients' rights.

- ❑ Raised awareness about the ethical responsibilities of researchers to obtain explicit, informed consent from patients before using their biological samples (tissue/cells) for research.
- ❑ Emphasized the importance of educating individuals about their rights and the implications of donating biological samples for research purposes.

Case Studies in Biotech Ethics: CRISPR Babies Case in China

- The CRISPR babies case refers to the controversial experiment in China where scientist He Jiankui claimed to have edited the genomes of twin girls (pseudonyms: "Lulu" and "Nana") to make them HIV-resistant. Experiment involved using CRISPR-Cas9 technology to edit the CCR5 gene, a gene that produces protein required by HIV to infect human cells.
- The case caused widespread ethical concerns and criticism within the scientific community due to the lack of transparency, oversight, and adherence to ethical guidelines.
- Editing the human embryo raised significant ethical dilemmas as changes made could be heritable and passed down to future generations, impacting the gene pool.
- The announcement of the CRISPR babies raised fears of unintended genetic consequences, off-target mutations, and unforeseen health risks associated with gene editing.

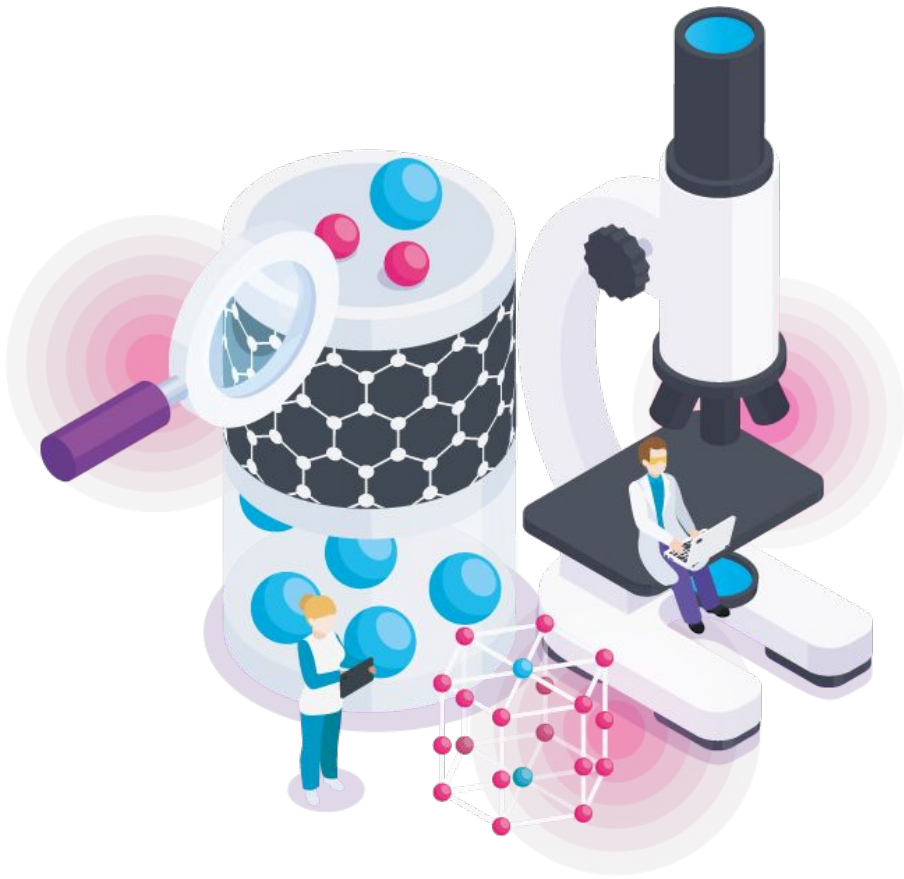
Biotech Ethics: Genetic Engineered Babies

Advantages

- Genetic screening reduces the baby's chances of being born with a serious genetic disease
- Increases the likelihood of a healthy baby reduces chances of miscarriage
- Can be used to save lives

Disadvantages

- Moral and ethical concerns:
 - Controlling nature
- Social concerns:
 - High-cost leads to gap in society
 - Genetically enhanced people may start to feel superior to those who haven't been enhanced
- Safety concerns:
 - Can lead to ovarian hyper-stimulation syndrome (OHSS)
 - Unforeseen health risks



THANK YOU

